Brief on Correlations of Sapiens

A Formal Toy Theory of Labels, Geometry, Dynamics, and Affinity

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Abstract

We develop a deliberately schematic mathematical language for describing labels, presentation, contextual transformations, and directed relationships between abstract agents. The point is to distinguish statements that follow from axioms from statements that require observations, semantics, or ethical judgments. After constructing an agent space, we prove a non-identifiability lemma, characterize when two social transformations commute, model four-agent and n-agent affinity networks by flag complexes, study a growing vocabulary of labels, and state conditional results concerning durable mutual affinity and conservation under social mixing. The theory is an exercise in formal modeling: none of its discrete encodings purports to classify actual human sex, gender, or relationships exhaustively.

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1 Scope, conventions, and the agent space

The original expression

$$\text{agent} := (\text{sex} \in 2^{\{0,1\}},\ \text{gender} \in 2^{S},\ \text{geometry} \subset \widehat{S}) \tag{1.1}$$

is best interpreted as a *typed schematic definition*, not as a statement about the number or nature of human categories. We now assign precise types to all three coordinates.

Definition 1.1 (Abstract agent)**.** *Fix the following mathematical data:*

(i) a finite code alphabet $E = \{0, 1\}$ *and its power set* $B = 2^E = \{\varnothing, \{0\}, \{1\}, \{0, 1\}\}$*;*

(ii) a nonempty set $\mathcal{G}$ *of descriptive tags, with label configurations* $\mathcal{A} = 2^{\mathcal{G}} = \mathcal{P}(\mathcal{G})$*;*

(iii) a nonempty smooth manifold M *(possibly with boundary), used solely as an abstract space of observable* presentation coordinates*.*

The agent space *and its projection maps are*

$$\boxed{\mathcal{X} := B \times \mathcal{A} \times M, \qquad x = (b, A, p), \qquad \pi_B(x) = b,\ \pi_{\mathcal{A}}(x) = A,\ \pi_M(x) = p.} \tag{1.2}$$

In particular, $|B| = 4$ is a theorem about this *encoding*. It is not a theorem that biological sex has precisely four states. Similarly, a tag in $\mathcal{G}$ is a symbol supplied by the modeler, not an objective diagnosis or an exhaustive description of personal identity. The power set permits multiple tags or no tags; it does not impose a binary taxonomy on gender.

Remark 1.2 (Context is a separate variable). A relationship or observation generally depends on context $c \in \mathcal{C}$, where $\mathcal{C}$ may encode time, social setting, communication history, and sampling conditions. We do not conflate c with the intrinsic coordinates of x. An optional fuller state is $(x, c) \in \mathcal{X} \times \mathcal{C}$.

1.1 Topology and the correction to the surface analogy

Give B the discrete topology. If $\mathcal{G}$ is finite, give $\mathcal{A}$ the discrete topology as well, and equip $\mathcal{X}$ with the product topology. If $\mathcal{G}$ is infinite, identify $2^{\mathcal{G}}$ with $\{0, 1\}^{\mathcal{G}}$ and equip it with the product topology. Neither construction requires the ill-typed quotient/exponent appearing in the preliminary notes.

Alternatively, when uncertainty about an *observer's assignment* of tags is of interest, one may introduce a probability distribution $q \in \mathsf{Prob}(\mathcal{G})$ and the usual weak topology on the space of probability measures. For finite $\mathcal{G}$ this is simply the Euclidean topology on the probability simplex

$$\Delta(\mathcal{G}) = \left\{ (q_g)_{g \in \mathcal{G}} : q_g \geq 0,\ \sum_{g \in \mathcal{G}} q_g = 1 \right\}. \tag{1.3}$$

Such a q models epistemic uncertainty; it need not represent an individual's identity as a mixture.

The preliminary "five-hole torus" language can be made precise as a *chosen example*, not a universal model of anatomy or identity. For the compact orientable surface $\Sigma_{g,r}$ of genus g with r boundary components,

$$\chi(\Sigma_{g,r}) = 2 - 2g - r. \tag{1.4}$$

Thus $\chi(\Sigma_{1,5}) = -5$, whereas $\chi(S^2) = 2$. The two surfaces are not homeomorphic. Moreover, *homotopy*, *homeomorphism*, *diffeomorphism*, and a *group action* are distinct notions; the expression $H \sim f_a$ defines none of them. No assertion about a "blow-up" or non-immersibility follows from (1.4).

2 Transformations, morphisms, and label changes

Definition 2.1 (Contextual action)**.** *A contextual transformation is a specified map* $f_c : \mathcal{X} \to \mathcal{X}$ *for* $c \in \mathcal{C}$*. A family of such maps is a monoid action only when there is a monoid* $(\mathcal{C}, \circ, e)$ *for which*

$$f_e = \mathrm{id}_{\mathcal{X}}, \qquad f_{c \circ d} = f_c \circ f_d. \tag{2.1}$$

A group action additionally requires invertible transformations.

For the discrete label component, it is often more appropriate to use an *admissible transition relation*

$$T_c \subseteq \mathcal{X} \times \mathcal{X}, \qquad x \xrightarrow{c} y \iff (x, y) \in T_c, \tag{2.2}$$

because a transition may be one-to-many or may not have an inverse. The word "homomorphism" should be reserved for a specified algebraic structure and maps preserving its operations; without those choices, (2.2) supplies arrows, not homomorphisms.

2.1 Example: the "Femboy Girl lemma"

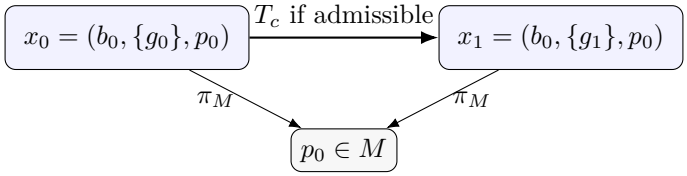
We retain the manuscript's name as a label for a mathematical thought experiment. It is not a claim about the attributes of any real group. Choose distinct tags $g_0, g_1 \in \mathcal{G}$, a presentation point $p_0 \in M$, and the same encoding $b_0 \in B$ at two states:

$$\begin{aligned} x_0 &= (b_0, \{g_0\}, p_0), \\ x_1 &= (b_0, \{g_1\}, p_0), \\ x_0 &\xrightarrow{c} x_1 \quad \text{if } (x_0, x_1) \in T_c. \end{aligned} \tag{2.3}$$

This repairs the unfinished "≡" in the source. The symbol $g_0 \to g_1$ describes a *possible modeled transition*; it does not follow from a visible feature or from the binary code.

Lemma 2.2 (Femboy Girl / non-identifiability lemma)**.** *If $|\mathcal{G}| \geq 2$, the projection $\pi_M : \mathcal{X} \to M$ is not injective. In particular, neither the tag configuration nor the code is uniquely determined by the presentation coordinate.*

Proof. Use the states in (2.3). Since $g_0 \neq g_1$, we have $x_0 \neq x_1$, but $\pi_M(x_0) = p_0 = \pi_M(x_1)$. Hence π_M is not injective. The same argument with distinct $b \in B$ proves non-identifiability of the code from p. □



Remark 2.3 (On "more homomorphisms"). Allowing more arrows in T_c increases the number of *admissible transitions* by definition. It does not establish that some real-life social class possesses more homomorphisms, nor that such transitions must occur. This distinction prevents a formal conclusion from being smuggled in as an empirical premise.

3 Correlations in the probabilistic sense

Let $(\Omega, \mathcal{F}, \mu)$ be a probability space and $X = (B_*, A_*, P_*)$ an $\mathcal{X}$-valued random variable. The statistical relationship between the coordinate variables is encoded in their *joint law*, not merely in the product $B \times \mathcal{A} \times M$.

For finite $B, \mathcal{A}$, write

$$p_{bA} = \mathbb{P}(B_* = b, A_* = A), \quad p_b = \sum_A p_{bA}, \quad p_A = \sum_b p_{bA}. \tag{3.1}$$

The mutual information (in nats) is

$$I(B_*; A_*) = \sum_{\substack{b,A \\ p_{bA}>0}} p_{bA} \log \frac{p_{bA}}{p_b p_A} = \mathrm{D}_{\mathrm{KL}}(\mu_{B_* A_*} \,\|\, \mu_{B_*} \otimes \mu_{A_*}) \geq 0. \tag{3.2}$$

Equality holds exactly when the two discrete variables are independent. This is a consequence of the nonnegativity of relative entropy.

Example 3.1 (A two-code, two-tag toy distribution). Suppose a distribution is concentrated on two code values and two tag configurations, with joint table

	A_0	A_1	marginal
b_0	0.4	0.1	0.5
b_1	0.1	0.4	0.5
marginal	0.5	0.5	1

Then

$$I(B_*; A_*) = 0.8\log(1.6) + 0.2\log(0.4) \approx 0.192745 \text{ nats}. \tag{3.3}$$

This numerical correlation is a property of the assumed table; it proves neither a causal link nor a universal correlation in humans.

If P_* takes values in a continuous manifold, correlations involving its law can likewise be studied using conditional distributions or relative entropy when the required measures are defined. There is no general scalar "correlation of geometry" without a chosen statistic.

4 Directed affinity, reciprocity, and commutativity

Take $n \geq 2$ agents $x_1, \ldots, x_n \in \mathcal{X}$. For a fixed time t and context c, define a directed affinity matrix

$$W(t,c) = (w_{ij}(t,c))_{1\leq i,j\leq n}, \quad w_{ij} \in [0,1], \quad w_{ii} = 0. \tag{4.1}$$

The entry w_{ij} is a modeling input, *not* a measured quantity until an actual measurement protocol is specified. Let $a_{ij}(t,c) \in \{0,1\}$ denote whether i has granted the particular modeled interaction toward j; neither affinity nor mutuality overrides this separate consent variable. Set

$$r_{ij}(t,c) = a_{ij}a_{ji}\min\{w_{ij}, w_{ji}\}. \tag{4.2}$$

Then $r_{ij} = r_{ji}$ even when $w_{ij} \neq w_{ji}$.

Lemma 4.1 (Conditional commutativity of independent changes)**.** *Suppose $F : U \to U$ and $H : V \to V$ are maps. Define $\widetilde{F}(u,v) = (F(u), v)$ and $\widetilde{H}(u,v) = (u, H(v))$ on $U \times V$. Then*

$$\widetilde{F} \circ \widetilde{H} = \widetilde{H} \circ \widetilde{F}. \tag{4.3}$$

Proof. Both compositions take (u,v) to $(F(u), H(v))$. □

The source's shorthand "$[bg, gb]$" and "$[gl, bl]$" has at least two possible meanings: exchange of labels on a *relationship*, or the commutator of *transformations*. These must not be confused. If a relationship is defined by r_{ij} in (4.2), exchanging i and j is symmetric. For arbitrary changes F and H of the same state, however, no commutativity theorem is available.

Example 4.2 (Noncommuting label permutations). On the three-state set $\{1,2,3\}$, let $F = (12)$ and $H = (23)$ be transpositions. Then

$$(F \circ H)(1) = 2, \qquad (H \circ F)(1) = 3. \tag{4.4}$$

Therefore $F \circ H \neq H \circ F$. If these are represented by linear permutation operators, their Lie bracket $[F,H] = FH - HF$ is nonzero. A group-theoretic commutator, when appropriate, instead uses $FHF^{-1}H^{-1}$.

Proposition 4.3 (Reciprocity is not directed symmetry)**.** *Symmetry of $R = (r_{ij})$ follows from (4.2) for every W and a. Symmetry of W itself is an independent hypothesis.*

Proof. Exchange i and j in (4.2); both the product $a_{ij}a_{ji}$ and the minimum are unchanged. □

5 From four-love to n-love: combinatorial topology

The topological construction concerns an *affinity network*, not the intrinsic identity of its vertices. Suppress t, c and fix a threshold $\theta \in (0,1]$. Define the undirected mutual-affinity graph G_θ by putting $\{i,j\}$ in its edge set exactly when $r_{ij} \geq \theta$. Its *clique (flag) complex* is

$$\mathcal{K}_\theta = \{\sigma \subseteq \{1,\ldots,n\} : \{i,j\} \in E(G_\theta) \text{ for all distinct } i,j \in \sigma\}. \tag{5.1}$$

Every complete $(k+1)$-vertex subgraph becomes a k-simplex.

Theorem 5.1 (Four-cycle to full four-way clique)**.** *Suppose the edge set of G_θ consists exactly of the cycle*

$$E(G_\theta) = \{\{1,2\},\{2,3\},\{3,4\},\{4,1\}\}.$$

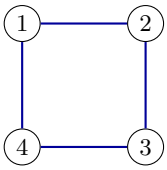
Then $\mathcal{K}_\theta$ has

$$H_1(\mathcal{K}_\theta;\mathbb{Z}) \cong \mathbb{Z}, \qquad \chi(\mathcal{K}_\theta) = 0. \tag{5.2}$$

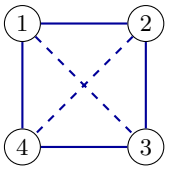
If all six edges are present instead, $\mathcal{K}_\theta$ is a three-simplex, and

$$H_1(\mathcal{K}_\theta;\mathbb{Z}) = 0, \qquad \chi(\mathcal{K}_\theta) = 1. \tag{5.3}$$

Proof. In the first case the flag complex is the one-dimensional cycle C_4, homeomorphic to S^1; thus $H_1 \cong \mathbb{Z}$ and $\chi = 4 - 4 = 0$. In the second case the clique complex consists of a full tetrahedron together with all its faces. A simplex is contractible, so its first homology vanishes and its Euler characteristic is $4 - 6 + 4 - 1 = 1$. □



C_4: a one-dimensional hole



K_4: flag complex fills all faces and the interior

Example 5.2 (Explicit weights and a topological transition)*.* Let all consent entries for distinct pairs equal 1 and take

$$W = \begin{pmatrix} 0 & .9 & .2 & .9 \\ .9 & 0 & .9 & .2 \\ .2 & .9 & 0 & .9 \\ .9 & .2 & .9 & 0 \end{pmatrix}. \tag{5.4}$$

At $\theta = 0.8$, only the cycle edges are present, so $\beta_1 = \operatorname{rank} H_1 = 1$. At $\theta = 0.2$, the two diagonals enter; all four vertices form a simplex, and $\beta_1 = 0$. The abrupt jump occurs in the *thresholded model* and should not be mistaken for a physical singularity in anyone's emotions.

Proposition 5.3 (n-cycle versus n-clique)**.** *For $n \geq 4$, the flag complex of the chordless cycle C_n has $H_1 \cong \mathbb{Z}$ and $\chi = 0$. The flag complex of the complete graph K_n is an $(n-1)$-simplex and has $H_1 = 0$ and $\chi = 1$.*

Proof. A chordless n-cycle for $n \geq 4$ has no 3-cliques, hence its flag complex is homeomorphic to a circle. A complete graph has every possible clique and therefore generates a simplex. □

This offers a precise meaning for the manuscript's phrase "four-love to n-love singularities": critical parameter levels where the homotopy type or Betti numbers of $\mathcal{K}_\theta$ change. A genuinely dynamical bifurcation would require a separate equation of motion and appropriate nondegeneracy assumptions.

6 Changing vocabularies: the "Europian" lemma

The preliminary notes anticipated future change in the number or meaning of tags. We retain "Europian" as an internal nickname only; it has no geographical or sociological content.

Definition 6.1 (Vocabulary filtration)**.** *A vocabulary filtration is a sequence of finite sets*

$$\mathcal{G}_0 \subseteq \mathcal{G}_1 \subseteq \mathcal{G}_2 \subseteq \cdots, \qquad \mathcal{A}_t = 2^{\mathcal{G}_t}, \qquad \mathcal{X}_t = B \times \mathcal{A}_t \times M. \tag{6.1}$$

The canonical inclusion sends $(b, A, p) \in \mathcal{X}_t$ to the same triple viewed as an element of $\mathcal{X}_{t+1}$.

Lemma 6.2 ("Europian" vocabulary-expansion lemma)**.** *If $|\mathcal{G}_{t+1}| = |\mathcal{G}_t| + k$, the number of possible discrete label configurations changes by a factor 2^k:*

$$|\mathcal{A}_{t+1}| = 2^k |\mathcal{A}_t|. \tag{6.2}$$

The canonical map $\mathcal{X}_t \hookrightarrow \mathcal{X}_{t+1}$ is injective.

Proof. For a finite set with m elements, its power set has 2^m elements. Inclusion preserves each triple exactly, so distinct triples remain distinct. □

Remark 6.3 (Expansion is not a demographic prediction). Equation (6.2) counts formal subsets. It does not predict that a population's observed identities will double, or that adding a term creates a previously nonexistent kind of person.

A separate issue arises with temporal continuity. If $\mathcal{G}$ is finite and label configurations carry the discrete topology, a continuous curve $A : [0, 1] \to 2^{\mathcal{G}}$ must be constant: the continuous image of the connected interval is connected, and connected subsets of a discrete space are singletons. Therefore changes of labels must be modeled using (i) jumps, (ii) a varying vocabulary, or (iii) an auxiliary continuous epistemic variable $q(t) \in \Delta(\mathcal{G})$. This is a mathematical obstruction to the initial informal idea that a discrete tag system can also vary continuously without modification.

Hypothesis 6.4 (Optional social evolution law). Given row-stochastic matrices K_t indexed by $\mathcal{A}_t$, a probability row vector ρ_t over label configurations can evolve as

$$\rho_{t+1} = \rho_t K_t, \quad K_t(A, A') \geq 0, \quad \sum_{A'} K_t(A, A') = 1. \tag{6.3}$$

This is an *assumed* Markov model, not a consequence of the agent space. If $\mathcal{G}_t$ changes, K_t must be rectangular with the old and new state spaces as its row and column indices.

7 Ideal love as an explicitly conditional definition

The mathematical system by itself cannot determine what "true love" means. Such a conclusion requires criteria chosen *outside* the algebra. We can nevertheless formalize one operational notion of a stable, mutual, consensual relationship.

Definition 7.1 (Stable mutual affinity on an interval)**.** *Fix a threshold $\theta \in (0, 1]$ and interval $I = [t_0, t_0 + T]$. A pair (i, j) is called θ-stable on I if*

$$a_{ij}(t) = a_{ji}(t) = 1 \quad \textit{and} \quad \min\{w_{ij}(t), w_{ji}(t)\} \geq \theta \quad \textit{for every } t \in I. \tag{7.1}$$

This is a stipulated criterion, not a psychological or moral theorem.

Theorem 7.2 (A local stability certificate)**.** *Assume consent holds throughout I and both directed affinities are L-Lipschitz in time: for $k\ell = ij, ji$,*

$$|w_{k\ell}(t) - w_{k\ell}(s)| \leq L|t - s|. \tag{7.2}$$

If $r_{ij}(t_0) \geq \theta + \varepsilon$ for $\varepsilon > 0$, then the pair is θ-stable on every interval $[t_0, t_0 + T]$ with $T \leq \varepsilon/L$, provided $L > 0$. If $L = 0$, it remains stable for the whole consent interval.

Proof. For $t \geq t_0$, the Lipschitz bounds give $w_{ij}(t) \geq w_{ij}(t_0) - L(t - t_0)$ and the analogous bound for w_{ji}. Taking the minimum yields

$$\min\{w_{ij}(t), w_{ji}(t)\} \geq r_{ij}(t_0) - L(t - t_0) \geq \theta + \varepsilon - LT \geq \theta. \tag{7.3}$$

Together with the maintained consent assumptions, this proves (7.1). □

Remark 7.3 (Limits of the certificate). This is a *sufficient local condition*, not a prediction of durable relationships. A withdrawal of consent invalidates the premise instantly, irrespective of the affinities. Emotional well-being, autonomy, honesty, reciprocity, and personal meaning cannot be established from a scalar w_{ij} without explicit, independently justified definitions.

8 Fuzzy–lazy dating: incomplete data and conservation

8.1 Underdetermination is not overdetermination

In an n-agent network there are $d = n(n-1)$ unknown directed non-diagonal affinities at a fixed time. Stack them into $z \in [0,1]^d$. A simplified observation model is

$$y = Hz + \eta, \qquad H \in \mathbb{R}^{m \times d}, \tag{8.1}$$

where y contains observations and η accounts for noise.

Proposition 8.1 (Underdetermined observations)**.** *If $\eta = 0$, $m < d$, and H has rank r, the unconstrained solution set of $Hz = y$, when nonempty, is an affine space of dimension $d - r$. If one feasible solution lies in the interior of $[0,1]^d$, there are infinitely many feasible solutions whenever $r < d$.*

Proof. For any particular solution z_0, the solutions form $z_0 + \ker H$. The rank–nullity theorem gives $\dim \ker H = d - r$. A sufficiently small neighborhood of an interior z_0 remains inside $[0,1]^d$, so nonzero kernel directions generate infinitely many feasible points. □

Example 8.2 (Three agents with two observations)*.* For $n = 3$ there are six directed variables. Knowing $w_{12} = 0.8$ and $w_{21} = 0.7$ leaves four variables unspecified. One cannot conclude whether $(2, 3)$ is reciprocally linked at threshold 0.6. Thus the unseen part of the network is *underdetermined*, not "overdetermined".

In contrast, an *overdetermined* system has more independent or redundant measurements than unknown parameters; it may be inconsistent under noise. A modeler may then minimize, for example,

$$\widehat{z} = \operatorname*{arg\,min}_{z \in [0,1]^d} \left\{ \frac{1}{2} \|Hz - y\|_2^2 + \frac{\lambda}{2} \|Lz\|_2^2 \right\}, \tag{8.2}$$

with a specified regularizer L and $\lambda \geq 0$. The regularizer supplies an assumption; it does not magically recover information absent from y.

8.2 A precise uniform-conservation statement

Let $u(t) \in \mathbb{R}^n$ describe a generic transferable quantity, such as attention units in a mathematical allocation model. (Real human affection need not obey a conservation law.) Consider the discrete evolution

$$u(t+1) = Pu(t), \qquad P_{ij} \geq 0. \tag{8.3}$$

Theorem 8.3 (Uniform conservation under doubly stochastic mixing)**.** *If P is* doubly stochastic*, meaning*

$$P\mathbf{1} = \mathbf{1}, \qquad \mathbf{1}^\mathsf{T} P = \mathbf{1}^\mathsf{T}, \tag{8.4}$$

then both the total and the uniform mean are invariant:

$$\boxed{\mathbf{1}^\mathsf{T} u(t) = \mathbf{1}^\mathsf{T} u(0), \qquad \overline{u}(t) = \frac{1}{n} \sum_{i=1}^{n} u_i(t) = \overline{u}(0).} \tag{8.5}$$

Proof. By (8.4), $\mathbf{1}^\mathsf{T} u(t+1) = \mathbf{1}^\mathsf{T} P u(t) = \mathbf{1}^\mathsf{T} u(t)$. Induction proves total conservation, and dividing by n proves mean conservation. □

Example 8.4 (Why row stochasticity alone is insufficient)*.* The matrix

$$P = \begin{pmatrix} 0.9 & 0.1 \\ 0.6 & 0.4 \end{pmatrix} \tag{8.6}$$

is row stochastic but not column stochastic. With $u(0) = (1, 0)^\mathsf{T}$, one obtains $u(1) = (0.9, 0.6)^\mathsf{T}$ and the total increases from 1 to 1.5. Conversely, $Q = \left(\begin{smallmatrix} 0.8 & 0.2 \\ 0.2 & 0.8 \end{smallmatrix}\right)$ is doubly stochastic and preserves the total.

Corollary 8.5 (Continuous-time network Laplacian)**.** *Let L be a graph Laplacian satisfying $\mathbf{1}^\top L = 0$, and let $\dot{u} = -Lu$. Then $\mathbf{1}^\top u(t)$ is constant.*

Proof. Differentiate: $\frac{d}{dt}(\mathbf{1}^\top u) = -\mathbf{1}^\top L u = 0$. □

9 Synthesis: what can and cannot be derived

The formal dependencies in this paper can be summarized as

$$\begin{gathered}(B, \mathcal{G}, M) \longrightarrow \mathcal{X} \longrightarrow (\text{state maps, observations}),\\ (W, a, \theta) \longrightarrow G_\theta \longrightarrow \mathcal{K}_\theta \longrightarrow H_k(\mathcal{K}_\theta),\\ \text{extra dynamical axioms} \longrightarrow \text{conditional stability and conservation statements.}\end{gathered} \tag{9.1}$$

The arrows in (9.1) record logical dependence, not causal relationships in the world.

1. A finite code or vocabulary gives a combinatorial state count. Its semantic completeness is *not* guaranteed.
2. Presentation need not identify labels. This is a precise non-injectivity assertion, not a psychological theory.
3. Affinity symmetry may be imposed by reciprocal aggregation; unrestricted transformations need not commute.
4. The topology of a thresholded graph can change with the threshold, giving a legitimate but model-dependent notion of "singularity."
5. Evolving vocabularies produce an inclusion system, not independently verifiable forecasts of social development.
6. "Ideal love" requires chosen criteria, and stability follows only under maintained assumptions such as consent and a Lipschitz bound.
7. Incomplete observations produce non-identifiability. Conservation requires a specific evolution operator.

Conclusion. The useful mathematical core is not a purported classification of *Homo sapiens*; it is the separation of *types, topology, observations, dynamics, and proofs.* Once these are distinguished, several formerly speculative statements become valid conditional lemmas. The remaining claims should be regarded as questions about empirical modeling, meaning, and values rather than deductions from symbolic notation.

A Notation and types

Symbol	Mathematical meaning
$E,\ B = 2^E$	Two-symbol alphabet and its four-element power set; formal encoding only.
$\mathcal{G},\ \mathcal{A} = 2^{\mathcal{G}}$	Tag vocabulary and allowable subsets of tags.
$M,\ \mathcal{X}$	Presentation manifold and product agent space $B \times \mathcal{A} \times M$.
$\mathcal{C},\ f_c,\ T_c$	Contexts; specified state maps; permitted state transitions.
$\chi,\ H_k$	Euler characteristic and singular homology groups.
$W = (w_{ij})$	Directed, context-dependent affinity weights in $[0, 1]$.
$a_{ij},\ r_{ij}$	Consent indicator and symmetrized mutual-affinity score.
$G_\theta,\ \mathcal{K}_\theta$	Threshold graph and its clique/flag complex.
$\mathcal{G}_t,\ \rho_t,\ K_t$	Time-indexed vocabulary, probability law, and stochastic evolution kernel.
H, z, y	Observation matrix, unknown affinity vector, and observed data.
P, L, u	Mixing matrix, network Laplacian, and generic conserved-state vector.

B Open questions for a genuinely empirical extension

A future empirical study would first need a defensible sampling frame, operational definitions of every recorded feature, consent to collect sensitive observations, uncertainty calibration, and a pre-specified statistical objective. Only then would questions such as the following be testable:

1. Which finite-dimensional features, if any, are adequate for a particular prediction task, and with what generalization error?
2. Does a chosen social transition kernel fit longitudinal observations better than a static baseline?
3. How sensitive are topological summaries of affinity networks to measurement noise and the threshold θ?
4. Under which conditions is a proposed stability measure robust to missing data, time variation, and asymmetric reports?
5. Can one test a conservation law against observations, rather than imposing it through a doubly stochastic transition matrix?

No empirical answers are assumed.